Karan Shah

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Education

Stony Brook University — *Stony Brook, NY*

MS in Computer Science | GPA : 3.64/4 — *Aug. 2019 - **Exp. Dec. 2020***

Coursework: Big Data Analytics, Theory of Database Systems, Machine Learning, Computer Vision, Data Science, Probability and Statistics

Gujarat Technological University — *Ahmedabad, India*

B.E. in Computer Engineering — *Aug. 2014 - May 2018*

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Skills

IDEs	Visual Studio, NetBeans, Jupyter Notebook, Eclipse, IntelliJ IDEA, Vim, Sublime
Programming	Python, Javascript, JAVA, C/C++, Golang
Tools/Technologies	AWS, MapReduce, Spark, Hadoop, Docker, Jenkins, Kubernetes, Git, SQL, NoSQL, Linux, Matlab
Frameworks/Libraries	NodeJS, Django, Flask, Angular, ElectronJS, Scikit-Learn, Numpy, Pandas, PyTorch, Tensorflow, Keras, OpenCV

Work Experience

Stony Brook University — *Stony Brook, NY*

Graduate Teaching Assistant — *January 2020 - May 2020*

- Assisted Prof. Dimitris Samaras in teaching Course CSE327-01: **Computer Vision**
- Managed assignment formulations and assessments, and resolve queries of students enrolled in the course

Simform Solutions — *Ahmedabad, India*

Software Engineer — *June 2018 - April 2019*

- Developed and automated **ETL** jobs to work on a huge realtime data using AWS services (Glue, Lambda, S3) and Spark
- Developed an end to end **Video Streaming system**; able to upload, scan and host the content on the cloud, and stream on cross-platform desktop application over globally separated clients; using **AWS services (Lambda, S3, Cloudfront, MQTT), NodeJS, and Python**
- Developed software system to semi-automate Company's marketing work with Chrome Extension and **REST** API Web backend using **Python and NodeJS**, efficient in reducing $\sim$ **40 %** of Marketing team's manual work

ISRO - Indian Space Research Organization — *Ahmedabad, India*

Research Intern, Machine Learning — *July 2017 - May 2018*

- Designed generalized **CNN** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for data preparation/processing of geographic remote sensing data, visualization, real-time training performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Projects

Child Mortality Analysis, Under Supervision of Prof. Andrew Schwartz (Python, Dask, Spark) — *April 2020 - May 2020*

- Analysed secondary factors contributing to child mortality through Big Data Techniques
- Implemented **Linear Regression and Similarity Search** running on women lifestyle real data distributed on Hadoop clusters

Retail Sales Data Analysis, Under Supervision of Prof. Steven Skiena (Python, Pandas) — *Oct. 2019 - Dec. 2019*

- Worked on a market basket dataset of a local chain retail stores to provide insights to help the company with their business
- Analysed data to find reasons for drop in sales with additional help of external datasets
- Implemented **Word2Vec and LSTM model with k-means clustering** to improve Shelf Space Management and Inventory management

Video Action Classification and Recognition (Python, PyTorch) — *Nov. 2019 - Dec. 2019*

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (inclass 3rd rank) on image test data, and **84%** (inclass 4th rank) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

Company's Internal Communication System (Java, Javascript) — *Sept. 2018 - Dec. 2018*

- Created a web-based software for employees of a Software Company to send/receive messages, share files in person, groups and broadcast channels with various administrative permission roles
- Utilised **SQLite** to store data of each user with **Java** Servlets to query the database and applied Software Engineering approaches

Desktop Application to Synchronize Local Storage with Cloud (AWS, Node.js, Electron.js) — *March 2017 - Apr. 2017*

- Built cross-platform application leveraging the ElectronJS framework
- Used **Node.js and AWS services** to fully synchronize local storage and AWS S3 storage

Publication

Band-wise Independent Pansharpening Using Convolutional Neural Networks with Shared Weights

Pankaj Bodani, Karan Shah, Shashikant Sharma

- Submitted to **IEEE** Geoscience and Remote Sensing Letters